HW 11

Due Tuesday, July 14 at 11:59pm.

AI use

Refer to the syllabus for our AI use policy. You will be asked to include any AI prompts used in this assignment at the end.

Starting a new Quarto file

For this assignment you will begin a new Quarto file. To do this, go to RStudio and click File > New File > Quarto Document. A pop-up will appear asking for a title, author, and format. You may leave these as the default options for now. Click OK and a new file will open. Update the YAML with your title, name, and date.

Data

We'll be revisiting the licorice example presented in HW 9. We'll assume that we can treat pain score as a continuous numeric variable. The data are available with the following code:

```r
library(tidyverse)
# read in data
licorice <- read.csv("data/licorice.csv")
```

Some relevant variables of interest are:

- `preOp_sex`: Sex (0 = Male; 1 = Female)
- `preOp_calcBMI`: Body mass index in kg/m2
- `preOp_asa`: The American Society of Anesthesiology severity score (1 = healthy patient, 2 = mild systemic disease, 3 = severe systemic disease)
- `treat`: Treatment given (0 = Sugar placebo; 1 = Licorice solution)
- `pacu30min_throatPain`: Sore throat pain score 30 minutes after arrival in PACU (11-point scale: 0 = No pain; 10 = Worst pain)
- `pacu90min_throatPain`: Sore throat pain score 90 minutes after arrival in PACU
- `postOp4hour_throatPain`: Sore throat pain score 4 hours after surgery

Exercises

In Exercise 1, use the `licorice` dataset. You may assume that the pain scores are continuous numeric variables for the purposes of this assignment.

Exercise 1

For this exercise, use the original `licorice` dataset.

We'd like to comprehensively evaluate whether the mean throat pain score 30 minutes after surgery (`pacu30min_throatPain`) varies by ASA classification.

a. What are the three assumptions of ANOVA?

b. Create a boxplot of ASA classification (x-axis) by `pacu30min_throatPain` (y-axis). Suppress warnings with `#| warning: false`. Based on the boxplot, do the data seem to be normally distributed within each group?

c. You decide to perform a Kruskal-Wallis test here. State the null and alternative hypotheses in words.

d. Use R to perform the Kruskal-Wallis test to evaluate whether the mean throat pain score 30 minutes after surgery varies by ASA classification. You should show code and output.

e. State your p-value, whether you reject or fail to reject the null hypothesis, and state your conclusion.

Exercise 2

For this exercise, refer to the test outlined in Lecture 11 Slide 8:

a. What is the name of this test?

b. List the steps to conducting this test.

c. Which assumption does this test NOT require that the paired t-test does require?

d. Does this test take into account the magnitude of the differences between paired values?

Cereal

The file `cereals.csv` contains data for 77 breakfast cereals in terms of their nutritional content per serving and other characteristics. The variables are:

- `name`: name of the cereal

- `mfr`: manufacturer (A: American Home Food Products, G: General Mills, K: Kellogg's, N: Nabisco, P: Post, Q: Quaker Oats, and R: Ralston Purina)

- `type`: cold vs. hot

- `calories`, `protein`, `fat`, `sodium`, `fiber`, `carbo`, `sugars`, `potass`: the amount of each of these nutrients, per gram, per serving (sodium and potassium are measured in milligrams)

- `shelf`: where the cereal was physically located (cereals on low aisles may be targeted toward children, for instance).

For the purposes of this homework, you may assume that this is a random sample of cereal types. Download `cereal.csv` from the Canvas Assignment page. Save it in your `data` folder within your project folder. You may load the dataset with the code below: (Ensure that you suppress any warnings and messages.)

```r
cereal <- read.csv("data/cereal.csv")
```

Exercise 3

For this exercise, we'll be investigating the research question: Is there evidence that there is a relationship between cereal manufacturer and where on the shelf their cereals are placed?

a. Use the following code to print the table of cereal placement versus manufacturer. (Yes, just copy and paste the code into your document.)

```r
table(cereal$shelf, cereal$mfr)
```

In general, are cell counts lower or higher compared to ideal counts for a Chi-Square test?

b. You decide to conduct a Fisher's Exact Test. State the null and alternative hypotheses in words.

c. Use R to conduct the test, displaying the code and output. Use $\alpha = 0.05$ To conduct this test, simply use `fisher.test(_____)`, and fill in the blank with the table code from part 4 (a).

d. State your p-value, whether you reject or fail to reject the null hypothesis, and state your conclusion.

AI Attestation

Was AI used on this assignment? If so, describe your prompts below. If not, simply state "AI was not used on this assignment."

Submission

As you've seen previously, we can **Render** into an .html file that can be opened by any web browser. To export it as a .pdf, open the file in your web browser and then print to or save as a .pdf document. Contact me in Ed Discussion if you need help!

You will submit the PDF documents for labs and homework to Gradescope as part of your final submission.

To submit your assignment:

- Access Gradescope through the menu on the BIOS 600 Canvas site.
- Click on the assignment, and you'll be prompted to submit it.
- Mark the pages associated with each exercise. All of the pages of your lab should be associated with at least one question (i.e., should be "checked").
- Select the first page of your .PDF submission to be associated with the "Formatting" section.

Grading

Component	Points
Ex 1	6
Ex 2	2
Ex 3	3
Formatting	3

The "Formatting" grade is to assess the document format. This includes having a neatly organized document (no excessive output, warnings/messages when loading packages and/or data) with readable code and your name and the date updated in the YAML.